

Olatomiwa Aluko

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EDUCATION

University of Central Florida (UCF)

Orlando, FL

Bachelor of Science in Computer Engineering

Expected May 2027

- Organizations: Colorstack, National Society of Black Engineers, Knight Hacks, Alpha Phi Alpha Fraternity, Inc.

EXPERIENCE

Bank of New York (BNY)

May 2026 – Aug 2026

Software Engineering Intern

Lake Mary, FL

- Shipped an **8-agent SDLC** review system catching architecture violations pre-review, cutting senior-engineer rework **2.5 hours per merge request** and saving **500+ engineering hours**.
- Engineered a **Python** CLI pipeline that used **LLMs** to map **5,000+ vendor financial fields** to BNY's canonical ontology, reducing mapping time **90%+** with parallel execution across **50+ workers**.
- Reproduced an external partner's **2,900-entity, 5,000-relationship** reference graph across two vendors in code, emitting output BNY's internal **knowledge graph API** could ingest directly instead of routing it through a transformation step.

National Society of Black Engineers – UCF Chapter

Nov 2025 – Present

Founding Software Engineer

Orlando, FL

- Architected and lead a **team of 4 engineers** building a full-stack engagement platform serving **100+ members** across **50+ annual events** with **RBAC** and real-time QR check-in.
- Reduced manual attendance processing **75%** by optimizing **PostgreSQL** queries and caching for **1,000+ concurrent** check-in operations.
- Hold **99.9% uptime** on a Dockerized **NestJS/Next.js** app deployed across **Railway** and **Vercel**, secured with **Google/Discord OAuth** and **JWT** access control.

Handshake

Oct 2025 – Present

Software Engineering Fellow

Remote

- Strengthened infrastructure measuring **AI model performance** across **10+ software engineering benchmark tasks** by reconstructing golden reference implementations and **fail-to-pass test suites** from real open-source **GitHub pull requests**.
- Enabled reproducible, isolated benchmark runs by provisioning **Docker-containerized** evaluation environments and commit workflows that cleared **automated CI checks** before every submission.

PROJECTS

Helix | *Go, Python, React, Kubernetes, NATS JetStream, PostgreSQL, ClickHouse*

- Built a runtime for running AI agents across a fleet of machines instead of one process, with a **Go** control plane handing work to **Python** workers over **NATS JetStream** so a dead worker never drops a job. Retraining the retrieval ranker on hard examples mined from failed runs lifted **Recall@10 35% (0.253 to 0.343)**.

Chalk | *Python, React, FastAPI, PostgreSQL/TimescaleDB, Redis, XGBoost*

- Built a site that forecasts how an NBA player will perform in a given game, training a separate **XGBoost/LightGBM** model per stat on **147,000+** past player-games. Over **1,070** live predictions it averaged **4.93 points of error** and hit accuracy targets on **5 of 7** stats against Vegas lines.

Aegis | *Go, Adaptive Concurrency Limiting, Circuit Breaking, Prometheus/Grafana*

- Built a **Go** reverse proxy that keeps a streaming service standing during traffic surges by capping in-flight work, dropping low-priority requests first, and cutting off failing backends. Under a **1,500 RPS** spike it served **11x more critical video segments** than the unprotected backend (**56.0% vs. 4.9%**) at **1.8ms p50**.

StandIn | *Python, Multi-Agent Orchestration, Intent Routing, LLM APIs*

- Won **Best Use of Auth0 AI Agents** at **LA Hacks** against nearly **1,000 participants** by designing the **multi-agent orchestrator** and intent-routing layer that classified user requests and dispatched them to specialized agents.

AWARDS

- **4x Hack Winner:** Microsoft x NSBE Agent-a-thon - **1st Place Overall**; LAHacks - **Best Use of Auth0 AI Agents**; HackPrinceton - **Runner-up, Regeneron Best AI & Tech for Clinical Trials**; RenderATL - **Best Use of Snowflake**.

SKILLS

Languages: Python, Java, Go, C/C++, TypeScript, SQL

Frameworks: React, Next.js, FastAPI, NestJS, Prisma, Jest, pytest

Infrastructure & Data: Kubernetes, Docker, PostgreSQL, Redis, NATS JetStream, ClickHouse, Supabase, Railway

Tools & Practices: Git, CI/CD, Prometheus, Grafana, OpenTelemetry, API Design, Unit/Integration Testing

AI & Automation: Claude Code, Codex, Cursor, Agentic SDLC, Multi-Agent Orchestration, MCP, LLM APIs